

BayRC Reference Manual

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BayRC-package

BayRC: Bayesian Framework for Comparative Circadian Genomics

Description

BayRC (Bayesian Rhythmicity Comparison) is a unified statistical framework for comparing and interpreting circadian rhythms across biological conditions: age, disease state, tissue, species, or sex. It jointly infers gene-level rhythmicity and phase, computes posterior probabilities of rhythmic and phase concordance, and classifies rhythmic gain, loss, conservation, and direction-specific phase shifts under Bayesian false discovery rate (BFDR) control. It further supports pathway-level enrichment and genome-wide concordance scoring, providing a unified, uncertainty-aware framework for comparative circadian analysis across tissues, species, and disease contexts.

Main functions

MCMC core (paper Sec. 2.1)

- `CB_init_single`
- `CBt_init_single`
- `CB_MCMC_single_rj_slice`
- `CB_getAllEst`
- `CBt_sim_data`
- `Cosinor_fit`
- `circular_HDI`
- `circular_median`

Single-group biomarker detection with BFDR control (paper Sec. 2.2)

- `match_symbols`
- `bfdr_from_posterior`
- `summarize_bay`

Two-group transition classification and phase concordance (paper Sec. 2.2)

- `detect_rhy`
- `transition_classify`

- `phase_infer`

Pathway-level rhythmic enrichment (paper Sec. 2.3)

- `pathSelect`
- `plot_heatmap`
- `multi_conservation_pathway`
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Genome-wide concordance summary (paper Sec. 2.4)

- `multi_conservation`
- `pairwise_concordance`

Cross-species alignment

- `match_homologs`
- `merge_mcmc`

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CB_init_single

Initialize single-species BayRC MCMC chain

Description

Produces starting values for the BayRC MCMC sampler by fitting an ordinary least-squares cosinor model to every gene. OLS-derived MESOR, amplitude, phase, and residual variance are used as initial values; rhythmicity is initialised from an F-test at $\alpha = 0.05$. When `FitCosinor = FALSE`, parameters are drawn from their priors.

Usage

```
CB_init_single(
  Data.list = list(data = as.data.frame(a.sim.dat$data[[1]]$dat), time =
    a.sim.dat$data[[1]]$x$time, gname = paste0("G",
    seq_len(nrow(a.sim.dat$data[[1]]$dat)))),
  P = 24,
  FitCosinor = TRUE,
  mu_M = 0,
  sigma_M = 10,
  mu_A = 1,
  sigma_A = 10,
  seed = 15213
)
```

Arguments

<code>Data.list</code>	A named list with three elements: <code>data</code> (G x N data.frame of expression values), <code>time</code> (length-N numeric Zeitgeber time vector in hours), and <code>gname</code> (length-G character vector of gene identifiers).
<code>P</code>	Numeric; circadian period in hours (default 24).
<code>FitCosinor</code>	Logical; if TRUE (default) use OLS cosinor estimates; if FALSE draw from priors.
<code>mu_M</code>	Numeric; prior mean for MESOR (default 0).
<code>sigma_M</code>	Numeric; prior standard deviation for MESOR (default 10).
<code>mu_A</code>	Numeric; prior mean for amplitude (default 1).
<code>sigma_A</code>	Numeric; prior standard deviation for amplitude (default 10).
<code>seed</code>	Integer; random seed (default 15213).

Details

Initialize MCMC chain for a single circadian dataset

Value

A named list with elements `rho` (length-G logical), `M` (length-G MESOR), `A` (length-G amplitude), `phi` (length-G acrophase in hours), and `sigma` (length-G residual variance). Suitable for passing directly to `CB_MCMC_single_rj_slice` as `Init.value`.

Examples

```
## Not run:
sim <- CBt_sim_data()
dat <- list(data = as.data.frame(sim$data[[1]]$dat),
           time = sim$data[[1]]$x$time,
           gname = paste0("G", seq_len(nrow(sim$data[[1]]$dat))))
init <- CB_init_single(dat)

## End(Not run)
```

CBt_init_single	<i>Initialize time-error BayRC MCMC chain</i>
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Description

Produces starting values for the CBt MCMC sampler, which models Zeitgeber-time measurement error (`t_p`). Parameter initialisation follows the same OLS cosinor approach as `CB_init_single` but also initialises the time-error latent variable `t_p` to zero.

Usage

```
CBt_init_single(
  Data.list = list(data = as.data.frame(a.sim.dat$data[[1]]$dat), time =
    a.sim.dat$data[[1]]$x$time, gname = paste0("G",
    seq_len(nrow(a.sim.dat$data[[1]]$dat)))),
  P = 24,
  FitCosinor = TRUE,
  mu_M = 0,
  sigma_M = 10,
  mu_A = 1,
  sigma_A = 10,
  seed = 15213
)
```

Arguments

<code>Data.list</code>	A named list with three elements: <code>data</code> (G x N data.frame of expression values), <code>time</code> (length-N numeric Zeitgeber time vector in hours), and <code>gname</code> (length-G character gene identifiers).
<code>P</code>	Numeric; circadian period in hours (default 24).
<code>FitCosinor</code>	Logical; if TRUE (default) use OLS cosinor estimates; if FALSE draw from priors.
<code>mu_M</code>	Numeric; prior mean for MESOR (default 0).
<code>sigma_M</code>	Numeric; prior standard deviation for MESOR (default 10).
<code>mu_A</code>	Numeric; prior mean for amplitude (default 1).
<code>sigma_A</code>	Numeric; prior standard deviation for amplitude (default 10).
<code>seed</code>	Integer; random seed (default 15213).

Details

Initialize MCMC chain for a dataset with Zeitgeber-time uncertainty

Value

A named list with elements `rho`, `M`, `A`, `phi`, `sigma` (same as `CB_init_single`), plus `t_p` (length-N vector of initial time-error values, set to 0). Suitable for passing to `CBt_MCMC_single` as `Init.value`.

Examples

```
## Not run:
sim <- CBt_sim_data()
dat <- list(data = as.data.frame(sim$data[[1]]$dat),
  time = sim$data[[1]]$x$time,
  gname = paste0("G", seq_len(nrow(sim$data[[1]]$dat)))
init <- CBt_init_single(dat)

## End(Not run)
```

CB_MCMC_single_rj_slice

CB MCMC single reversible-jump slice sampler

Description

Runs the BayRC RJMCMC sampler for a single-species/condition circadian expression dataset. Within each iteration the sampler proposes model-space jumps (rho: 0/1 rhythmicity indicator) via an RJMCMC acceptance ratio and updates amplitude, phase, MESOR, and residual variance via Gibbs or slice steps.

Usage

```
CB_MCMC_single_rj_slice(
  Data.list,
  Init.value,
  P = 24,
  iteration = 3000,
  thin = 20,
  n.burn = 1000,
  seed = 15213,
  diagnostics = FALSE,
  p_rhythmic = rep(0.2, 100),
  rj.p.stay = 0.5,
  A_prior = "Jeffreys_OLS_condi",
  mu_A = 1,
  sigma_A = 10^2,
  A.min = 0,
  A_wb_beta2 = 2,
  A_gm_shape = 1.99,
  A_gm_rate = 0.5,
  rj.phi = TRUE,
  rj.A = TRUE,
  mu_M = 0,
  sigma_M = 10^2,
  sigma_prior_v = 4,
  sigma_prior_s = 1,
  save.file = "MCMC_save.rds",
  save.file2 = "MCMC_save.rds"
)
```

Arguments

Data.list	A named list with three elements: data (G x N data.frame of expression values), time (length-N numeric Zeitgeber time vector in hours), and gname (length-G character vector of gene identifiers).
Init.value	List returned by CB_init_single containing starting values for rho, M, A, phi, and sigma.
P	Numeric; circadian period in hours (default 24).
iteration	Integer; total number of MCMC iterations including burn-in (default 3000).

<code>thin</code>	Integer; thinning interval; every <code>thin</code> -th post-burn-in sample is stored (default 20).
<code>n.burn</code>	Integer; number of burn-in iterations discarded before storing samples (default 1000).
<code>seed</code>	Integer; random seed for reproducibility (default 15213).
<code>diagnostics</code>	Logical; if <code>TRUE</code> , compute and print basic post-run convergence diagnostics (RJMCMC acceptance rate, mean effective sample size for rho) and attach them as <code>\$diagnostics</code> on the returned list. Warns if mean ESS for rho falls below 100 (default <code>FALSE</code>).
<code>p_rhythmic</code>	Numeric vector of length <code>G</code> ; prior probability $\Pr(\rho_g = 1)$ for each gene. Enters the RJMCMC log-prior odds as $\log(p / (1 - p))$. Default is <code>rep(0.2, 100)</code> .
<code>rj.p.stay</code>	Numeric in (0, 1); probability of skipping the between-model move (staying in the current model) at each iteration (default 0.5).
<code>A_prior</code>	Character; prior on amplitude. One of "Jeffreys_OLS_condi", "trunc_Normal", "Jeffreys", "Jeffreys_ridge", "sq_expo", or "gamma" (default "Jeffreys_OLS_condi").
<code>mu_A</code>	Numeric; mean of the truncated-Normal amplitude prior when <code>A_prior = "trunc_Normal"</code> (default 1).
<code>sigma_A</code>	Numeric; variance of the truncated-Normal amplitude prior (default 10^2).
<code>A.min</code>	Numeric; lower truncation bound for amplitude (default 0).
<code>A_wb_beta2</code>	Numeric; beta parameter for the squared-exponential amplitude prior <code>sq_expo</code> (default 2).
<code>A_gm_shape</code>	Numeric; shape parameter for the gamma amplitude prior; values ≤ 2 give a flatter prior (default 1.99).
<code>A_gm_rate</code>	Numeric; rate parameter for the gamma amplitude prior; smaller values give a flatter prior (default 0.5).
<code>rj.phi</code>	Logical; if <code>TRUE</code> , phase ϕ is jointly proposed in the RJMCMC birth/death step (default <code>TRUE</code>).
<code>rj.A</code>	Logical; if <code>TRUE</code> , amplitude A is jointly proposed in the RJMCMC birth/death step (default <code>TRUE</code>).
<code>mu_M</code>	Numeric; prior mean for MESOR (default 0).
<code>sigma_M</code>	Numeric; prior variance for MESOR (default 10^2).
<code>sigma_prior_v</code>	Numeric; degrees-of-freedom hyperparameter of the inverse-gamma prior on residual variance (default 4).
<code>sigma_prior_s</code>	Numeric; scale hyperparameter of the inverse-gamma prior on residual variance (default 1).
<code>save.file</code>	Character; path for saving intermediate MCMC output (default "MCMC_save.rds").
<code>save.file2</code>	Character; secondary save path used inside the RJMCMC helper (default "MCMC_save.rds").

Details

Run reversible-jump MCMC for a single circadian dataset

Value

A named list of $G \times K$ matrices ($K = \text{floor}(\text{iteration} / \text{thin})$):

rho Posterior samples of the rhythmicity indicator in $\{0, 1\}$.

phi Posterior samples of acrophase in hours (Zeitgeber scale).

A Posterior samples of amplitude.

M Posterior samples of MESOR.

sigma Posterior samples of residual variance.

log.r1, log.r1.SS, log.r3_A, log.r3_phi RJMCMC log-ratio components (for diagnostics).

if.accept.rj Binary matrix indicating accepted RJMCMC moves.

All matrices have rownames equal to `Data.list$gname`. After calling `match_symbols()`, the attributes `attr(rho, "symbols")` and `attr(rho, "RHYindex")` are set.

Examples

```
## Not run:
sim <- CBt_sim_data()
dat <- list(data = as.data.frame(sim$data[[1]]$dat),
           time = sim$data[[1]]$x$time,
           gname = paste0("G", seq_len(nrow(sim$data[[1]]$dat))))
init <- CB_init_single(dat)
mcmc_out <- CB_MCMC_single_rj_slice(dat, init,
                                   iteration = 200, thin = 10, n.burn = 100)

## End(Not run)
```

CB_getAllEst

Get all posterior estimates from BayRC MCMC output

Description

Computes posterior estimates and highest-density credible intervals for amplitude (A), acrophase (phi), MESOR (M), residual variance (sigma), and optionally peak time (t_p) from the MCMC output list. Circular statistics (`circular_median` and `circular_HDI`) are used for phi/t_p; linear posterior means and HDI are used for the rest.

Usage

```
CB_getAllEst(
  res,
  burn = 100,
  CI = TRUE,
  credMass = 0.95,
  P = 24,
  rhythmic = FALSE
)
```

Arguments

res	Named list; MCMC output from <code>CB_MCMC_single_rj_slice</code> (or <code>CBt_MCMC_single</code>). Must contain matrices <code>rho</code> , <code>A</code> , <code>phi</code> , <code>M</code> , <code>sigma</code> with <code>attr(rho, "symbols")</code> and <code>attr(rho, "RHYindex")</code> set by <code>match_symbols</code> .
burn	Integer; number of initial MCMC iterations to discard as additional burn-in when computing estimates (default 100).
CI	Logical; if <code>TRUE</code> (default), return lower and upper credible-interval bounds in addition to the point estimate.
credMass	Numeric in (0, 1); nominal coverage of the HDI (default 0.95).
P	Numeric; period in hours used for circular statistics (default 24).
rhythmic	Logical; if <code>TRUE</code> , return only rows for genes classified as rhythmic (<code>RHYindex == 1</code> ; default <code>FALSE</code>).

Details

Extract posterior estimates and credible intervals for all parameters

Value

A list of data.frames, one per parameter, each with columns `<param>.Est`, `<param>.Lower`, `<param>.Upper` and `RHYindex`. Row names are gene symbols when `attr(rho, "symbols")` is set.

Examples

```
## Not run:
sim <- CBt_sim_data()
dat <- list(data = as.data.frame(sim$data[[1]]$dat),
           time = sim$data[[1]]$x$time,
           gname = paste0("G", seq_len(nrow(sim$data[[1]]$dat))))
init <- CB_init_single(dat)
res <- CB_MCMC_single_rj_slice(dat, init,
                             iteration = 300, thin = 10, n.burn = 100)
est <- CB_getAllEst(res, burn = 10)

## End(Not run)
```

CBt_sim_data

Simulate CBt circadian dataset

Description

Generates synthetic multi-group circadian expression data following the BayRC cosinor model, optionally with Zeitgeber-time measurement error (`t_p`) on a specified proportion of samples. Each gene's true phase is drawn uniformly from $[0, P)$, and expression is generated as $Y = M + A * \cos(\omega * (t + t_p) - \phi) + \epsilon$. Supports multiple rhythmicity/phase-shift scenarios defined by `TOJR.grid`.

Usage

```
CBt_sim_data(
  P = 24,
  TOJR.grid = NULL,
  dParam = NULL,
  G = NULL,
  n = 30,
  fixed_tp = TRUE,
  sigma.t = 2,
  n.sigma.t = 16,
  p.sigma.t = NULL,
  Params = list(A = 0.5, phi = 0, M = 5, sigma = 1),
  TimePoints = NULL,
  PhaseClust = NULL
)
```

Arguments

P	Numeric; circadian period in hours (default 24).
TOJR.grid	Data.frame or matrix; G_scenario x n.group binary rhythmicity indicator grid defining which groups are rhythmic for each scenario.
dParam	Named list; group-level parameter offsets (dA, dPhase, dM) relative to the reference group, one element per non-reference group.
G	Integer vector of length n_scenario; number of genes per scenario row.
n	Integer; number of samples (default 30).
fixed_tp	Logical; if TRUE (default), time errors are fixed at +/- sigma.t; if FALSE, they are drawn from N(0, sigma.t).
sigma.t	Numeric; magnitude of the Zeitgeber-time measurement error (default 2).
n.sigma.t	Integer or NULL; number of samples with non-zero t_p. Exactly one of n.sigma.t and p.sigma.t must be non-NULL.
p.sigma.t	Numeric or NULL; proportion of samples with non-zero t_p.
Params	Named list; base cosinor parameters: A (amplitude), phi (phase), M (MESOR), sigma (residual SD).
TimePoints	Numeric vector or NULL; fixed observation times; if NULL, times are drawn uniformly from \[0, P).
PhaseClust	Currently reserved (default NULL).

Details

Simulate circadian expression data with Zeitgeber-time measurement error

Value

A named list with elements:

data A list of length n.group; each element contains x (sample-level covariates including time, time.p), beta (gene-level true parameters), and dat (G x N expression matrix).

TOJR Data.frame; expanded rhythmicity grid concatenated with parameter offsets for all genes.

Examples

```
## Not run:
sim <- CBT_sim_data(n.sigma.t = 8)

## End(Not run)
```

Cosinor_fit

Ordinary least-squares cosinor fit

Description

Fits a single-period OLS cosinor model ($y = M + A \cdot \cos(\omega \cdot t - \phi) + \epsilon$) to every gene in a dataset using vectorised least squares, returning MESOR, amplitude, acrophase, R-squared, and an F-test p-value for rhythmicity. This is the classical (non-Bayesian) cosinor method the paper contrasts BayRC's RJMCMC approach with; useful as a fast baseline or a quick pre-screen before running the full MCMC pipeline.

Usage

```
Cosinor_fit(
  x,
  period = 24,
  amp.cutoff = 0,
  p.adjust.method = "BH",
  parallel.ncores = 1
)
```

Arguments

<code>x</code>	A list with elements <code>data</code> (G x N data.frame or matrix), <code>time</code> (length-N numeric Zeitgeber time in hours), and <code>gname</code> (length-G gene names).
<code>period</code>	Numeric; circadian period in hours (default 24).
<code>amp.cutoff</code>	Numeric; amplitude cutoff (currently unused, reserved for future filtering; default 0).
<code>p.adjust.method</code>	Character; multiple-testing correction method passed to <code>p.adjust</code> (default "BH").
<code>parallel.ncores</code>	Integer; number of cores (currently reserved, default 1).

Details

Fit OLS cosinor model to circadian expression data

Value

The input list `x` augmented with a `rhythm` data.frame (one row per gene) containing columns `gname`, `M` (MESOR), `A` (amplitude), `phase / peak` (acrophase in hours), `R2`, `sigma`, `pvalue`, and `qvalue` (BH-adjusted p-value). Also adds `x$P`.

Examples

```
G <- 10; N <- 48
Y <- matrix(rnorm(G * N), G, N)
tv <- seq(0, 47, length.out = N)
x <- list(data = Y, time = tv, gname = paste0("G", 1:G))
res <- Cosinor_fit(x)
head(res$rhythm)
```

circular_HDI	<i>Compute the shortest circular highest-density interval</i>
--------------	---

Description

Finds the shortest interval on a circle of circumference P that contains at least `credMass` of the posterior samples. Uses a sliding-window algorithm on a doubled sorted array to handle wrap-around.

Usage

```
circular_HDI(samples, credMass = 0.95, P = 24, a = 0)
```

Arguments

<code>samples</code>	Numeric vector; phase samples in hours.
<code>credMass</code>	Numeric; desired coverage probability (default 0.95).
<code>P</code>	Numeric; period / circle circumference in hours (default 24).
<code>a</code>	Numeric; left endpoint of the normalisation interval (default 0).

Value

List with elements `lower` and `upper` (both in $\setminus[a, a + P)$) giving the bounds of the shortest HDI.

circular_median	<i>Compute the circular median of phase samples</i>
-----------------	---

Description

Converts samples from hours to radians, computes the circular median via `circular::median.circular`, and converts back to hours.

Usage

```
circular_median(samples, P = 24)
```

Arguments

<code>samples</code>	Numeric vector; phase samples in hours.
<code>P</code>	Numeric; period in hours (default 24).

Value

Numeric scalar; circular median in hours in $\backslash[0, P)$.

match_symbols	<i>Map Ensembl IDs to gene symbols and set rhythmicity attributes</i>
---------------	---

Description

Takes the raw list output from `CB_MCMC_single_rj_slice` and (optionally) converts Ensembl row names to HGNC gene symbols via biomaRt. If row names are already symbols they are used directly. Computes a Bayes Factor for each gene using `summarize_bay` and sets `attr(rho, "symbols")` and `attr(rho, "RHYindex")` so that downstream functions can access gene identifiers and rhythmicity classifications. Duplicate symbols are resolved by keeping the row with the highest Bayes Factor.

Usage

```
match_symbols(input_df, BF, p_rhythmic = 0.5, ensemble = NULL)
```

Arguments

input_df	Named list; raw MCMC output (e.g. from <code>CB_MCMC_single_rj_slice</code>). Must contain a <code>rho</code> matrix with gene identifiers as row names.
BF	Numeric; Bayes Factor threshold above which a gene is called rhythmic.
p_rhythmic	Numeric in (0, 1); prior probability of rhythmicity used to compute the Bayes Factor (default 0.5).
ensemble	A biomaRt Mart object; required when row names are Ensembl IDs (default NULL).

Details

Annotate MCMC output with gene symbols and rhythmicity index

Value

The input list filtered to unique symbols, with all matrix elements row-subsetted and `attr(*, "symbols")`, `attr(*, "RHYindex")`, and (if Ensembl IDs were present) `attr(*, "ensembl_gene_ids")` set on each matrix.

Examples

```
## Not run:
res <- CB_MCMC_single_rj_slice(dat, init)
res2 <- match_symbols(res, BF = 3, p_rhythmic = 0.2)

## End(Not run)
```

bfdr_from_posterior

BFDR threshold from posterior rhythmicity probabilities

Description

Implements the BFDR rule described in the BayRC paper (Eq. 25): sort genes by decreasing posterior probability p_g , compute the running average of $(1 - p_g)$, and return the largest threshold τ_c such that $\text{BFDR}(\tau_c) = \text{mean}(1 - p_g \mid p_g \geq \tau_c) \leq \alpha$. This controls the expected proportion of false rhythmic calls at level α .

Usage

```
bfdr_from_posterior(posterior_probs, alpha = 0.05)
```

Arguments

posterior_probs Numeric vector; marginal posterior probability of rhythmicity for each gene (values in $[0, 1]$).

alpha Numeric; desired BFDR level (default 0.05).

Details

Estimate the Bayesian False Discovery Rate threshold from posterior probabilities

Value

A list with elements:

threshold Numeric; the BFDR-optimal cutoff τ_c .

rhythmic_genes Logical vector; TRUE for called rhythmic.

n_rhythmic Integer; number of rhythmic calls.

bfdr_values Numeric vector; BFDR at each ranked position.

sorted_probs Sorted (descending) posterior probabilities.

detect_rhy

Detect rhythmic genes in two conditions using BFDR control

Description

Applies `bfdr_from_posterior` separately to the marginal posterior rhythmicity probabilities of two conditions and returns the BFDR-optimal rhythmic gene sets for each, together with data.frames of annotated gene information.

Usage

```
detect_rhy(dat1, dat2, bfdr_alpha = 0.05)
```

Arguments

`dat1` Named list; MCMC output for condition A with a `rho` matrix.
`dat2` Named list; MCMC output for condition B with a `rho` matrix.
`bfdr_alpha` Numeric; BFDR level (default 0.05).

Value

A list with elements:

rhythmic_A_logical, rhythmic_B_logical Logical vectors for conditions A and B.
rhythmic_A, rhythmic_B Data.frames of rhythmic genes with posterior probability columns.
threshold_A, threshold_B BFDR thresholds for each condition.
n_rhythmic_A, n_rhythmic_B, n_total, bfdr_alpha Counts and settings.

<code>summarize_bay</code>	<i>Summarise posterior rhythmicity and Bayes Factor per gene</i>
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Description

Computes the posterior mean rhythmicity (`RowAverage`), the number of zero samples (`Zeroes`), and the Bayes Factor for each gene. The Bayes Factor is computed as posterior odds / prior odds: $BF = (\text{rho_bar} / (1 - \text{rho_bar})) / (p_rhythmic / (1 - p_rhythmic))$. Genes with $BF >$ the threshold are called rhythmic (`Rhythmicity = 1`).

Usage

```
summarize_bay(input_df, BF, p_rhythmic = 0.2)
```

Arguments

`input_df` G x K binary matrix of posterior `rho` samples.
`BF` Numeric; Bayes Factor threshold for calling a gene rhythmic.
`p_rhythmic` Numeric; prior probability of rhythmicity (default 0.5).

Value

Data.frame with columns `Zeroes`, `RowAverage`, `Rhythmicity` (0/1), and `BayesF`.

```
transition_classify
```

Bayesian transition classification (gain / loss / maintained)

Description

Computes joint transition posterior probabilities: $p_{\text{gain}} = (1 - p_A) * p_B$ (gained rhythmicity), $p_{\text{loss}} = p_A * (1 - p_B)$ (lost rhythmicity), $p_{\text{cons}} = p_A * p_B$ (maintained rhythmicity), then applies `bfd_r_from_posterior` to each to identify gain, loss, and conserved gene sets at the specified BFDR level.

Usage

```
transition_classify(pA, pB, bfd_r_alpha = 0.05)
```

Arguments

<code>pA</code>	Numeric vector; marginal posterior rhythmicity probability for condition A (reference; length G).
<code>pB</code>	Numeric vector; marginal posterior rhythmicity probability for condition B (length G).
<code>bfd_r_alpha</code>	Numeric; BFDR control level (default 0.05).

Details

Classify rhythmicity transitions using joint BFDR on transition posteriors

Value

A list with elements:

gain_loss_status Named character vector of length G with values "Gain", "Loss", "Maintained", or "Non-rhythmic".

gain_genes, loss_genes, cons_genes Integer indices of genes in each class.

tau_gain, tau_loss, tau_cons BFDR thresholds for each transition type.

n_gain, n_loss, n_cons Counts.

results Data.frame with per-gene classification details.

phase_infer	<i>Bayesian phase inference with BFDR-controlled shift/conservation flags</i>
-------------	---

Description

For genes classified as "Maintained" by `transition_classify` or `transition_classify_marginal`, computes the posterior distribution of the phase difference $\Delta\phi = \phi_1 - \phi_2$ (wrapped to $[-P/2, P/2]$), estimates the probability of a significant shift ($|\Delta\phi| \geq \text{shift}$), and applies BFDR control to flag significantly shifted or conserved genes. Optionally computes the full circular HDI of $\Delta\phi$ when `compute_hdi = TRUE`.

Usage

```
phase_infer(
  phi_matrix1,
  phi_matrix2,
  gain_loss_status,
  P = 24,
  credMass = 0.95,
  shift = 2,
  a = -P/2,
  bfdr_alpha = 0.05,
  compute_hdi = FALSE
)
```

Arguments

<code>phi_matrix1</code>	G x K matrix; posterior phase samples for condition 1.
<code>phi_matrix2</code>	G x K matrix; posterior phase samples for condition 2.
<code>gain_loss_status</code>	Named character vector of length G with values "Gain", "Loss", "Maintained", or "Non-rhythmic" (from <code>transition_classify</code>).
<code>P</code>	Numeric; period in hours (default 24).
<code>credMass</code>	Numeric; HDI coverage (default 0.95); only used when <code>compute_hdi = TRUE</code> .
<code>shift</code>	Numeric; phase-shift threshold in hours (default 4).
<code>a</code>	Numeric; left endpoint for circular normalisation (default $-P/2$).
<code>bfdr_alpha</code>	Numeric; BFDR control level (default 0.05).
<code>compute_hdi</code>	Logical; if TRUE, compute the circular HDI for each maintained gene (slower; default FALSE).

Details

Infer phase shifts and conservation among maintained rhythmic genes

Value

A list with per-gene vectors:

peak1, peak2 Circular median phases for each condition.

deltaPhi.Est, deltaPhi.Lower, deltaPhi.Upper Phase-difference median and HDI (filled only when `compute_hdi = TRUE`).

prob_shift, prob_conserved Posterior probabilities of shift and conservation.

flag_shift, flag_cons, flag_undetermined BFDR-significant flags.

BFDR_shift, BFDR_cons Running BFDR values.

pathSelect

FGSEA-based pathway selection for circadian transitions

Description

Applies fast gene-set enrichment analysis (FGSEA) to a posterior gene-ranking metric derived from two BayRC MCMC outputs. Genes are ranked by their log-odds-ratio of gain vs loss, or by conservation, gain, loss, or union probability; pathways are tested for enrichment toward either end of this ranking. Pathway-level effect sizes and probabilistic gain/loss/conservation indices are added to the output.

Usage

```
pathSelect (
  mcmc.merge.list,
  pathway.list,
  dataset.names = NULL,
  ranking.method = c("log_odds_ratio", "union", "conserved", "gain", "loss"),
  score_type = c("std", "pos", "neg"),
  pathwaysize.lower.cut = 10,
  pathwaysize.upper.cut = 200,
  qvalue.cut = 0.05,
  epsilon = 0.001,
  n_top_genes = 5,
  nperm = 10000,
  nproc = 1,
  seed = 12345
)
```

Arguments

`mcmc.merge.list`

Named list of exactly 2 MCMC output lists; the first is condition A (reference) and the second is condition B. Each must have `attr(rho, "symbols")` set.

`pathway.list` Named list of character vectors; pathway gene sets.

`dataset.names`

Character vector of length 2 or `NULL`; labels for conditions A and B (default auto-detected from list names).

```

ranking.method      Character; gene-ranking metric: one of "log_odds_ratio" (default), "union",
                    "conserved", "gain", or "loss".
score_type          Character; fgsea score type: "std" (two-sided), "pos" (gain only), or "neg"
                    (loss only, inverts ranking).
pathwaysize.lower.cut      Integer; minimum pathway size (default 10).
pathwaysize.upper.cut      Integer; maximum pathway size (default 200).
qvalue.cut            Numeric; adjusted p-value significance cutoff (default 0.05).
epsilon              Numeric; small constant added to probabilities to avoid log(0) (default 0.001).
n_top_genes          Integer; number of top genes to report per category per pathway (default 5).
nperm                Integer; fgsea permutations (default 10000).
nproc                Integer; fgsea parallel processes (default 1).
seed                 Integer; random seed (default 12345).

```

Details

Select circadian-enriched pathways using FGSEA

Value

A list with elements:

results Data.frame of fgsea results (one row per pathway) augmented with Gain_Index, Loss_Index, Conserved_Index, Gain_Loss_Ratio_Arithmetic, expected counts, and top gene columns.

gene_rankings Data.frame of per-gene ranking metrics and posterior probabilities.

parameters List of analysis settings.

Examples

```

## Not run:
res <- pathSelect(list(human = human_res, mouse = mouse_res),
                  pathway.list = msig_pathways,
                  ranking.method = "log_odds_ratio")
head(res$results)

## End(Not run)

```

plot_heatmap

Pathway-level expression heatmap

Description

Produces a ComplexHeatmap visualisation for a given pathway, showing posterior rhythmicity probabilities and phase information for both conditions side by side. Genes can be coloured by gain/loss/ maintained status from transition_classify output. The plot can show all genes, only rhythmic genes, or both (two panels).

Usage

```
plot_heatmap(
  data1,
  data2,
  pathway_genes,
  pathway_name,
  phase_results,
  transition_results = NULL,
  group_names = c("Group1", "Group2"),
  save_path = NULL,
  n_bins = 24,
  versions = c("full", "rhythmic_only", "both")
)
```

Arguments

<code>data1</code>	Named list; MCMC output for condition 1 with <code>rho</code> and <code>phi</code> matrices.
<code>data2</code>	Named list; MCMC output for condition 2.
<code>pathway_genes</code>	Character vector; gene names in the pathway.
<code>pathway_name</code>	Character; pathway label used in the plot title.
<code>phase_results</code>	Output from <code>phase_infer</code> or a similar list with per-gene phase metrics.
<code>transition_results</code>	Output from <code>transition_classify</code> or <code>NULL</code> (default <code>NULL</code>); used to colour genes by gain/loss/maintained status.
<code>group_names</code>	Character vector of length 2; condition labels (default <code>c("Group1", "Group2")</code>).
<code>save_path</code>	Character or <code>NULL</code> ; file path for saving the plot PNG; if <code>NULL</code> the plot is drawn but not saved.
<code>n_bins</code>	Integer; number of bins for the phase colour wheel (default 24).
<code>versions</code>	Character; which version to produce: "full", "rhythmic_only", or "both" (default "full").

Details

Integrated pathway heatmap with rhythmicity and phase information

Value

Called for side effects; invisibly returns the heatmap object.

multi_conservation_pathway

Multi-condition pathway circadian conservation analysis

Description

Runs all pairwise pathway conservation comparisons among a list of MCMC outputs: computes observed pathway congruence scores, generates permutation null distributions, applies BH correction, and writes results to an Excel file.

Usage

```
multi_conservation_pathway(
  mcmc.merge.list,
  dataset.names,
  select.pathway.list,
  delta = 3,
  units = "hours",
  B = 1000,
  ncores = NULL,
  parallel = "auto",
  rounds = 1,
  save_intermediate = FALSE,
  intermediate_dir = "intermediate_results",
  output.dir = "Conservation_Pathway"
)
```

Arguments

<code>mcmc.merge.list</code>	Named list of MCMC output lists, one per condition.
<code>dataset.names</code>	Character vector; condition labels.
<code>select.pathway.list</code>	Named list of character vectors; pathway gene sets.
<code>delta</code>	Numeric; phase threshold (default 3).
<code>units</code>	Character; "hours" (default).
<code>B</code>	Integer; permutations per pair (default 1000).
<code>ncores</code>	Integer or NULL; parallel cores.
<code>parallel</code>	Logical or "auto".
<code>rounds</code>	Integer; rounds per pair.
<code>save_intermediate</code>	Logical; save intermediate results.
<code>intermediate_dir</code>	Character; directory for intermediate saves.
<code>output.dir</code>	Character; directory for Excel output (default "Conservation_Pathway").

Details

Pairwise pathway conservation analysis across multiple conditions

Value

Named list of pairwise pathway result data.frames. Also writes an Excel file to `output.dir`.

multi_conservation_pathway_bootstrap

Pathway-level bootstrap concordance (compatibility wrapper)

Description

Thin wrapper around `multi_conservation` retained for backward compatibility with analysis scripts that call `multi_conservation_pathway_bootstrap`. All computation is delegated to `multi_conservation`; the `delta` and `units` arguments are accepted but unused (the c-score is threshold-free by design; see paper Eq. 3).

Note: output Excel sheets are named "Results" and "Column_Definitions", not "congruence_index" as in the legacy Thien/ version. Update downstream `read.xlsx` calls accordingly.

Usage

```
multi_conservation_pathway_bootstrap(
  mcmc.merge.list,
  dataset.names,
  select.pathway.list,
  delta = 3,
  units = "hours",
  n_boot = 500,
  n_perm = 1000,
  output.dir = "Conservation_Pathway_Bootstrap",
  ...
)
```

Arguments

<code>mcmc.merge.list</code>	Named list of MCMC outputs.
<code>dataset.names</code>	Character vector of dataset labels.
<code>select.pathway.list</code>	Named list of pathway gene sets.
<code>delta</code>	Numeric; ignored (reserved for future phase-threshold concordance variant).
<code>units</code>	Character; ignored.
<code>n_boot</code>	Integer; bootstrap replicates (default 500).
<code>n_perm</code>	Integer; permutation replicates (default 1000).
<code>output.dir</code>	Character; directory for Excel output.
<code>...</code>	Additional arguments passed to <code>multi_conservation</code> .

Value

Same value as `multi_conservation`.

multi_conservation *Multi-dataset pairwise pathway concordance analysis*

Description

Runs all pairwise pathway bootstrap concordance analyses across a list of MCMC outputs and compiles results into Excel output. Wraps `bootstrap_conservation_pathway_within`.

Usage

```
multi_conservation(
  mcmc.merge.list,
  dataset.names,
  select.pathway.list,
  n_perm = 1000,
  n_boot = 500,
  alpha = 0.05,
  min.p = 5,
  qvalue_threshold = NULL,
  output.dir = "Conservation_Pathway_Bootstrap",
  save_output = TRUE,
  use_cpp = FALSE,
  compute_pvalue = TRUE,
  compute_ci = TRUE
)
```

Arguments

<code>mcmc.merge.list</code>	Named list of MCMC output lists.
<code>dataset.names</code>	Character vector; condition labels.
<code>select.pathway.list</code>	Named list of pathway gene sets.
<code>n_perm</code>	Integer; permutations per pathway (default 1000).
<code>n_boot</code>	Integer; bootstrap replicates per pathway (default 500).
<code>alpha</code>	Numeric; significance level (default 0.05).
<code>min.p</code>	Integer; minimum number of pathway genes required to test a pathway (default 5).
<code>qvalue_threshold</code>	Numeric or NULL; q-value significance cutoff.
<code>output.dir</code>	Character; directory for Excel output.
<code>save_output</code>	Logical; if TRUE (default), write results to an Excel file in <code>output.dir</code> . Set FALSE to skip the write (e.g. in tests or CI).
<code>use_cpp</code>	Logical; if TRUE, use the Rcpp-accelerated concordance computation (default FALSE, pure R).
<code>compute_pvalue</code>	Logical; if TRUE (default), run the permutation test for statistical significance.
<code>compute_ci</code>	Logical; if TRUE (default), run bootstrap resampling for a 95% confidence interval on the adjusted concordance.

Value

Named list of pairwise result data.frames. Also writes an Excel file to `output.dir`.

```
pairwise_concordance
```

Compute pairwise concordance matrix across multiple tissues

Description

For each pair of tissues in `human_data`, computes the Jaccard concordance index between their posterior rhythmicity classifications. Optionally restricts the comparison to the top `n_gene` genes ranked by mean posterior rhythmicity across all tissues.

Usage

```
pairwise_concordance(human_data, n_gene = NULL)
```

Arguments

<code>human_data</code>	Named list of MCMC output lists; each element must have a <code>rho</code> matrix with matching gene row names.
<code>n_gene</code>	Integer or <code>NULL</code> ; if specified, restrict to the top <code>n_gene</code> genes by mean rhythmicity across tissues.

Value

Square symmetric numeric matrix of Jaccard concordance values, with row and column names equal to `names(human_data)`.

```
match_homologs
```

Match homologs across species MCMC outputs

Description

Uses biomaRt to retrieve one-to-one orthologs between each non-reference species and the reference species (default human), then intersects the mapped gene sets across all datasets. Each input list is row-filtered and reordered so that all datasets share the same genes in the same order, enabling direct cross-species comparisons.

Usage

```
match_homologs(input_dfs, species_from, ref = "human")
```

Arguments

<code>input_dfs</code>	A list of MCMC output lists (one per dataset), each with matrices whose row names are Ensembl gene IDs and whose <code>attr(rho, "ensembl_gene_ids")</code> attribute is set.
<code>species_from</code>	Character vector of the same length as <code>input_dfs</code> ; species label for each dataset (currently supports "human" and "mouse").
<code>ref</code>	Character; reference species for the common gene space (default "human").

Details

Align multiple MCMC outputs to a common set of homologous genes

Value

A list of the same length as `input_dfs`, each element row-filtered to the intersection of homologous reference-space gene IDs and reordered consistently.

Examples

```
## Not run:
aligned <- match_homologs(list(human_res, mouse_res),
                           species_from = c("human", "mouse"))

## End(Not run)
```

merge_mcmc

Merge MCMC output lists by orthologs

Description

Intersects gene sets across a list of MCMC outputs from different species or conditions. When `ortholog.db` is supplied and multiple species are present, the ortholog database maps each species' genes to a common reference gene space (defined by the reference dataset). When datasets share a single species (or no ortholog database is provided), the function simply takes the intersection of gene names. If `uniqG = TRUE`, duplicate ortholog matches are resolved by keeping the entry with the highest mean absolute posterior signal.

Usage

```
merge_mcmc(mcmc.list, species, ortholog.db = NULL, reference = 1, uniqG = T)
```

Arguments

<code>mcmc.list</code>	A list of MCMC output matrices or lists, one per condition or study. Row names must be gene identifiers.
<code>species</code>	Character vector of the same length as <code>mcmc.list</code> ; species label for each dataset.
<code>ortholog.db</code>	Data.frame or matrix with one column per species (column names matching species); each row maps orthologous genes across species. If <code>NULL</code> , datasets are merged by gene name intersection only (default <code>NULL</code>).
<code>reference</code>	Integer; index of the reference dataset whose gene names define the merged row names (default 1).
<code>uniqG</code>	Logical; if <code>TRUE</code> (default), keep only the highest-signal match when multiple orthologs map to the same reference gene.

Details

Merge multiple MCMC outputs by matching orthologous genes

Value

A list of the same length as `mcmc.list`, each element row-subsetted and reordered to the common gene set, with row names from the reference dataset.

Examples

```
## Not run:
merged <- merge_mcmc(list(human_res, mouse_res),
  species      = c("human", "mouse"),
  ortholog.db  = hm_orth,
  reference    = 1)

## End(Not run)
```

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